

## **ASSIGNMENT #06**

**Course:** CHE 341: CHEMICAL PROCESS CONTROL  
**Term:** Spring 2026  
**Professor:** Dr. Michael Caracotsios  
**Office:** EIB 244  
**Cell Phone:** (630) 696-0934  
**E-mail:** [mcaracot@uic.edu](mailto:mcaracot@uic.edu)

**Name :** \_\_\_\_\_ Omar Zaruk \_\_\_\_\_

**Due Date:** Friday, April 17<sup>th</sup> TA's Mailbox

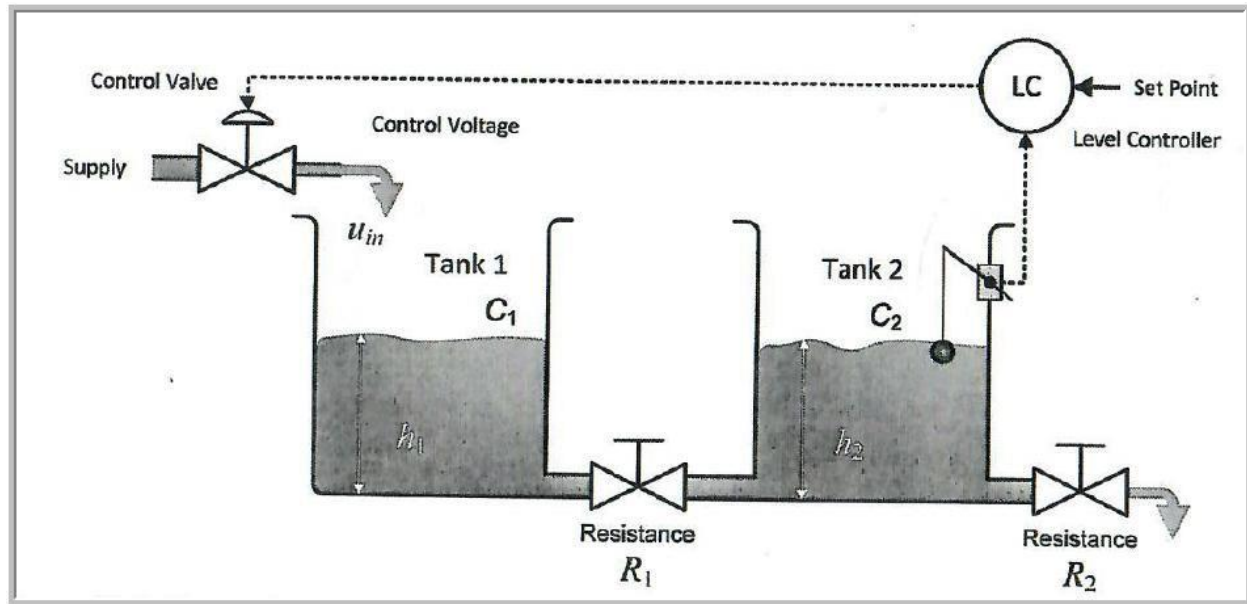
### **INSTRUCTIONS FOR ASSIGNMENT DELIVERY**

- Prepare a **typed** report of your assignment.
- Include this cover page with your name(s) in the report.
- Upload your report as a PDF file by the Due Date and Time on the UIC Blackboard.
- Place a hard copy of your report as a PDF file by the Due Date and Time in the TA's mailbox.
  
- Assignments that are not typed will not be accepted.
- Late Assignments will not be accepted without prior approval.
- You may use the Internet as well as ChatGPT or other public tools as long as you place a written statement about their use in your assignment report

## Chemical Process Control

### **Problem:** Two-Tank Level Control System

In many chemical process applications, cascaded tank systems are used to attenuate disturbances and improve level regulation. Consider the two-tank system shown below, where liquid flows from Tank 1 into Tank 2.



A level transmitter is installed in Tank 2 and provides feedback to a PID controller, which manipulates the inlet flow rate to regulate the level in Tank 2.

The capacitance of the first and second tank is  $C_1=C_2=1.0 \text{ m}^3/\text{m}$  and the outlet resistances of the first and second tank  $R_1=0.5 \text{ m}/(\text{m}^3/\text{min})$  and  $R_2=0.333 \text{ m}/(\text{m}^3/\text{min})$ . The volumetric flow rate through the gate valves varies linearly with the pressure drop. Assume that the time delays in the control loop including the transportation lag between the first and second tank amount to around 35% of the dominant time constant of the system.

1. Derive the process and disturbance transfer functions assuming that the inlet flow acts as the manipulated variable (MV) and also as the disturbance variable (DV).
2. Design a PID controller with  $\text{GM}=3$  and  $\text{PM}=60$  degrees and perform a simulation for set point and disturbance change of unit size.
3. Compare your controller with:
  - a. Ziegler-Nichols Open Loop
  - b. Cohen-Coon
  - c. Ziegler-Nichols Closed Loop
  - d. Luyben-Tyres
  - e. IMC-based tuning
  - f. Your own design

Present your results in a tabular form and compute the following metrics for both set point tracking and disturbance rejection of unit size.

1. Overshoot (if present)

2. Settling time
3. Gain Margin
4. Phase Margin
5. Maximum amount of time delay allowed before the onset of instability
6. Peak value  $M_s$  of the sensitivity function
7. Peak value  $M_t$  of the complementary sensitivity function

Include all the simulation graphs in your final report along with the Excel table shown below

|                                                         | GM=3<br>PM=60 deg | Ziegler-Nichols<br>Open Loop | Cohen-Coon | Ziegler-Nichols<br>Closed Loop | Luyben-Tyreus | IMC | My Personal<br>Controller | Selected<br>Controller |
|---------------------------------------------------------|-------------------|------------------------------|------------|--------------------------------|---------------|-----|---------------------------|------------------------|
| Overshoot (if present)                                  |                   |                              |            |                                |               |     |                           |                        |
| Settling time                                           |                   |                              |            |                                |               |     |                           |                        |
| Gain Margin                                             |                   |                              |            |                                |               |     |                           |                        |
| Phase Margin                                            |                   |                              |            |                                |               |     |                           |                        |
| Maximum amount of time delay allowed before instability |                   |                              |            |                                |               |     |                           |                        |
| Peak Value of the sensitivity function                  |                   |                              |            |                                |               |     |                           |                        |
| Peak Value of the complimentary sensitivity function    |                   |                              |            |                                |               |     |                           |                        |
|                                                         |                   |                              |            |                                |               |     |                           |                        |
| Controller Gain, $K_c$                                  |                   |                              |            |                                |               |     |                           |                        |
| Controller Reset Time, $\tau_i$                         |                   |                              |            |                                |               |     |                           |                        |
| Controller Derivative Time, $\tau_D$                    |                   |                              |            |                                |               |     |                           |                        |

Discuss the trade-offs observed between performance and robustness for the different controllers. Comment on which controller provides the best overall performance and justify your answer.

## 1. Dynamic Model and Transfer Function Development

The process considered in this assignment consists of two interacting tanks connected in series. Liquid enters the first tank, flows into the second tank through a hydraulic resistance, and finally exits the system through another resistance. The liquid level in the second tank is measured and used as the controlled variable. This configuration is widely used in industrial processes because it helps reduce disturbances in upstream equipment while maintaining stable downstream operation.

The objective of this section is to develop the mathematical model of the system and derive both the process and disturbance transfer functions. Since the inlet flow acts as both the manipulated variable and the disturbance variable, the inlet stream must be represented as the sum of these two contributions.

Let

$$\begin{aligned}h_1(t) &= \text{liquid level in Tank 1 (m)} \\h_2(t) &= \text{liquid level in Tank 2 (m)}\end{aligned}$$

The system parameters are given as

$$\begin{aligned}C_1 &= C_2 = 1.0 \text{ m}^3/\text{m} \\R_1 &= 0.5 \text{ m}/(\text{m}^3/\text{min}) \\R_2 &= 0.333 \text{ m}/(\text{m}^3/\text{min})\end{aligned}$$

The inlet flow is written as

$$q_{in}(t) = u(t) + d(t)$$

where

$$\begin{aligned}u(t) &= \text{manipulated inlet flow} \\d(t) &= \text{disturbance flow}\end{aligned}$$

### Flow Relationships

Because the valve flows vary linearly with pressure drop, the flow from Tank 1 to Tank 2 and the outlet flow from Tank 2 can be written as

$$q_{12}(t) = \frac{h_1(t) - h_2(t)}{R_1}$$

$$q_2(t) = \frac{h_2(t)}{R_2}$$

### Mass Balance on Tank 1

Applying a dynamic mass balance to Tank 1 gives

$$C_1 \frac{dh_1(t)}{dt} = q_{in}(t) - q_{12}(t)$$

Substituting the flow expressions gives

$$C_1 \frac{dh_1(t)}{dt} = u(t) + d(t) - \frac{h_1(t) - h_2(t)}{R_1}$$

### Mass Balance on Tank 2

Applying a dynamic mass balance to Tank 2 gives

$$C_2 \frac{dh_2(t)}{dt} = q_{12}(t) - q_2(t)$$

Substituting the flow expressions gives

$$C_2 \frac{dh_2(t)}{dt} = \frac{h_1(t) - h_2(t)}{R_1} - \frac{h_2(t)}{R_2}$$

These two differential equations form the complete dynamic model of the interacting two-tank system.

### Laplace Transform Equations

Taking the Laplace transform of both balances, assuming zero initial conditions, gives

$$\begin{aligned} C_1 s H_1(s) &= U(s) + D(s) - \frac{H_1(s) - H_2(s)}{R_1} \\ C_2 s H_2(s) &= \frac{H_1(s) - H_2(s)}{R_1} - \frac{H_2(s)}{R_2} \end{aligned}$$

Rearranging gives

$$\begin{aligned} \left(C_1 s + \frac{1}{R_1}\right) H_1(s) - \frac{1}{R_1} H_2(s) &= U(s) + D(s) \\ -\frac{1}{R_1} H_1(s) + \left(C_2 s + \frac{1}{R_1} + \frac{1}{R_2}\right) H_2(s) &= 0 \end{aligned}$$

Solving for  $H_2(s)$  gives the transfer function

$$G_p(s) = G_d(s) = \frac{2}{s^2 + 7s + 6}$$

or in factored form

$$G_p(s) = G_d(s) = \frac{2}{(s + 1)(s + 6)}$$

### Time Delay

The poles are located at

$$s = -1 \text{ and } s = -6$$

Therefore, the dominant time constant is

$$\tau_{dom} = 1 \text{ min}$$

The problem states that the total loop time delay is approximately 35% of the dominant time constant. Therefore,

$$\begin{aligned} \theta &= 0.35 \tau_{dom} \\ \theta &= 0.35 \text{ min} \end{aligned}$$

### Final Process Model

Including the time delay, the final transfer functions become

$$\begin{aligned} G_p(s) &= \frac{2e^{-0.35s}}{(s + 1)(s + 6)} \\ G_d(s) &= \frac{2e^{-0.35s}}{(s + 1)(s + 6)} \end{aligned}$$

These expressions describe a stable second-order process with time delay and provide the basis for controller design and comparison in the following sections.

## 2. Controller Design for GM = 3 and PM = 60°

The next step in the analysis was to design a feedback controller that satisfies specified robustness requirements. The required gain margin was 3 and the required phase margin was 60°. These values were selected because they provide a balanced compromise between responsiveness and stability while maintaining tolerance to modeling uncertainty and additional time delay.

A proportional–integral (PI) controller was selected for this part of the assignment. PI control is commonly used in level-control systems because it eliminates steady-state error while maintaining stable behavior. In addition, the design problem involved two unknown controller parameters, namely the controller gain and the integral time, making the PI structure appropriate for solving the robustness conditions.

### Controller Transfer Function

The controller was represented in standard PI form as

$$G_c(s) = K_c \left( 1 + \frac{1}{\tau_I s} \right)$$

where

$$\begin{aligned} K_c &= \text{controller gain} \\ \tau_I &= \text{integral time (min)} \\ \tau_D &= 0 \end{aligned}$$

Derivative action was omitted because it is generally less important for level-control processes and would introduce an additional parameter that was not required for satisfying the specified robustness targets.

### Process Parameters Used for Design

The process model derived in Section 1 was used directly for the controller design:

$$G_p(s) = \frac{0.333e^{-0.35s}}{(1.0s + 1)(0.1667s + 1)}$$

From this model, the process constants were identified as

$$\begin{aligned}
 K_p &= 0.333 \\
 \tau_1 &= 1.0 \text{ min} \\
 \tau_2 &= 0.1667 \text{ min} \\
 \theta &= 0.35 \text{ min}
 \end{aligned}$$

These constants describe the gain, time constants, and transportation delay of the interacting two-tank system.

### Determination of Controller Constants

The controller constants were determined using frequency-domain stability analysis. The Bode stability method was used to evaluate the open-loop response and identify the controller parameters required to satisfy the specified gain margin and phase margin.

The target conditions were

$$\begin{aligned}
 GM &= 3 \\
 PM &= 60^\circ
 \end{aligned}$$

Using the Bode stability analysis, the controller parameters were adjusted until both robustness conditions were satisfied simultaneously.

The resulting controller constants were

$$\begin{aligned}
 K_c &= 3.65 \\
 \tau_I &= 1.132 \text{ min} \\
 \tau_D &= 0
 \end{aligned}$$

These values produced the desired stability margins while maintaining stable closed-loop behavior.

### Closed-Loop Simulation

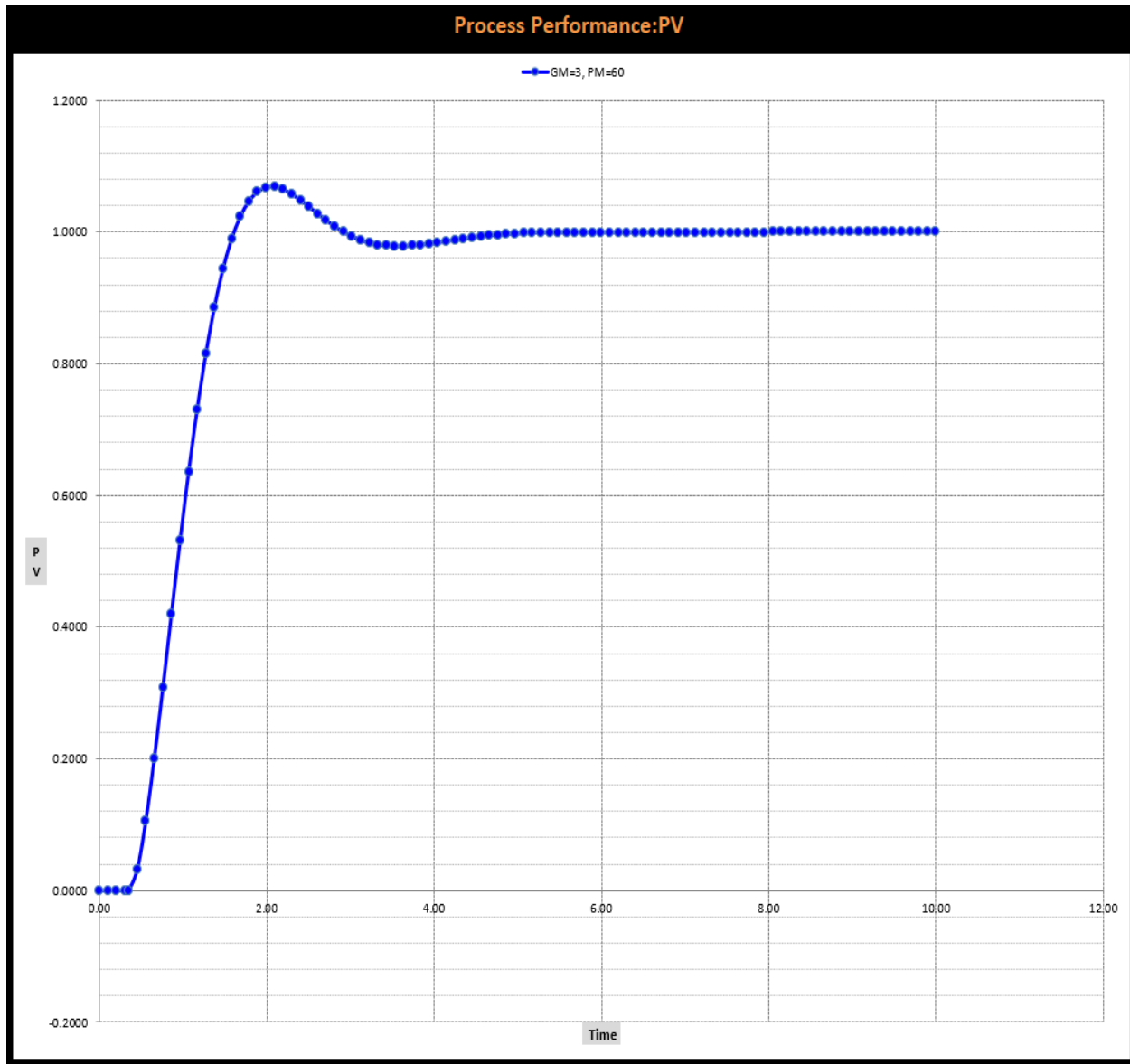
After the controller constants were determined, the time-domain response of the closed loop was evaluated using dynamic simulation. The same process model was retained during the simulation so that the results reflected the behavior of the actual system.

Two test cases were considered:

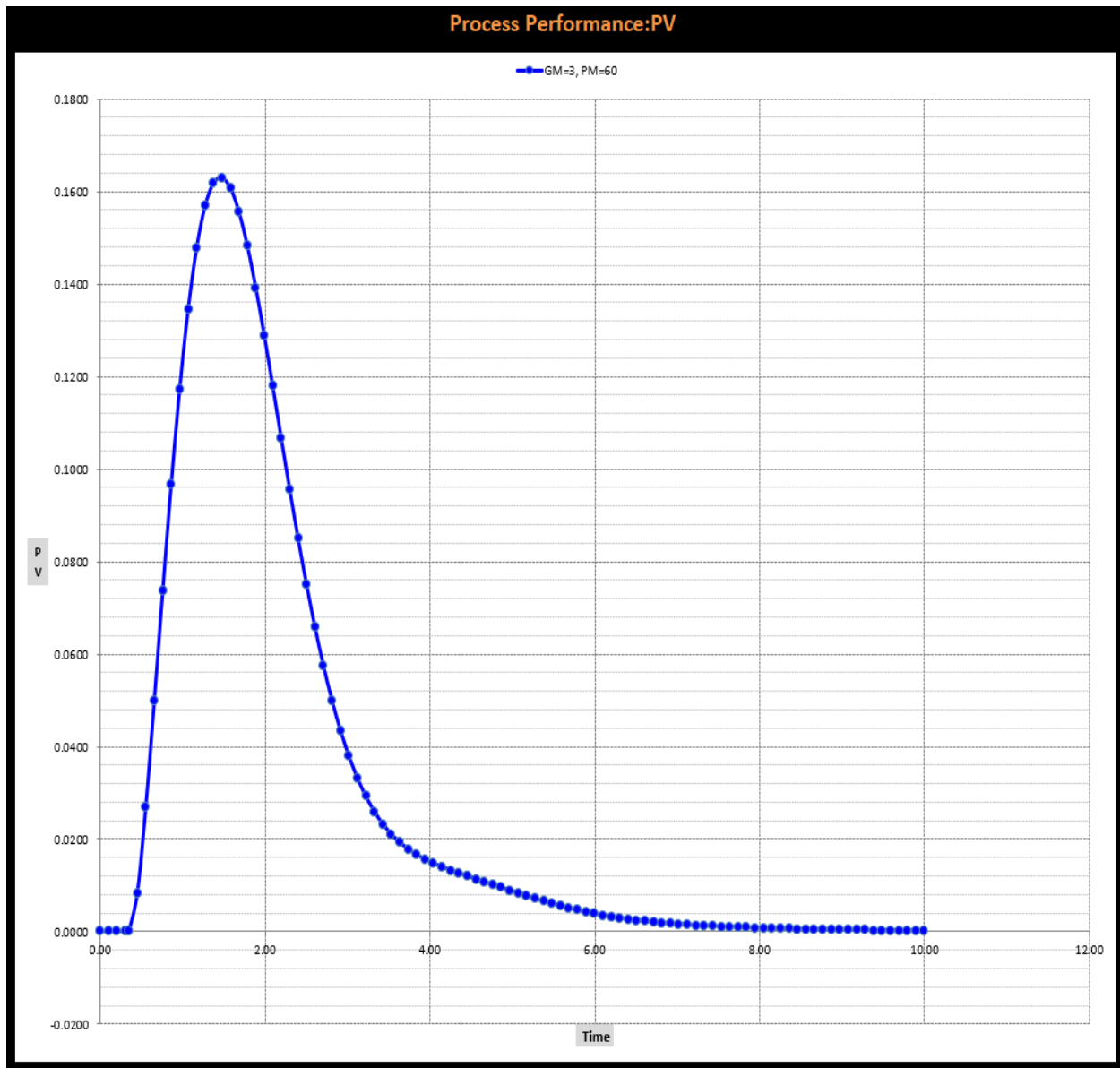
1. Unit set-point change
2. Unit disturbance change

In the set-point test, the desired liquid level was increased by one unit while the disturbance was held constant. In the disturbance test, the disturbance input was increased by one unit while the set point remained fixed.

The results showed that the controller produced a stable and well-damped response. The process variable approached the desired value with moderate overshoot and returned smoothly to steady state without sustained oscillation. In addition, the disturbance response demonstrated effective rejection of load changes, confirming that the controller maintained stable operation under both operating conditions.



*Figure 1- Closed-loop response for unit set-point change*



*Figure 2- Closed-loop response for unit disturbance change*

**Figure 2:** Closed-loop response for unit disturbance change

### 3. Comparison of Standard Tuning Methods and Selection of a Final Controller

After completing the controller design in Section 2, the next objective was to compare the performance of this controller with several standard tuning methods commonly used in industrial process control. The methods examined in this study included:

- Ziegler–Nichols Open Loop
- Cohen–Coon
- Ziegler–Nichols Closed Loop

- Tyreus–Luyben
- Internal Model Control (IMC)
- Personal controller design

The controller developed in Section 2 was retained as the reference case. Each controller was evaluated using the same plant model so that the comparison reflected differences in tuning rather than differences in process representation.

### 3.1 Controller Derivation Strategy

Not all controller constants were obtained in the same way. Some tuning methods required a first-order-plus-time-delay approximation of the process, while others were based directly on the ultimate gain and ultimate period or on specified robustness targets.

The following controllers required a first-order-plus-time-delay approximation:

- Ziegler–Nichols Open Loop
- Cohen–Coon
- IMC

These methods rely on simplified process models for calculating controller parameters.

The remaining controllers did not require process approximation:

- Ziegler–Nichols Closed Loop
- Tyreus–Luyben
- Personal Controller
- GM = 3, PM = 60° Controller

These controllers were determined either directly from the full process model or from selected tuning constants.

### 3.2 FOPTD Approximation Using Skogestad Rule

For the methods that required simplified process representation, the original second-order process was approximated by a first-order-plus-time-delay model using Skogestad's half-rule.

The original process constants were

$$\begin{aligned}\tau_1 &= 1.0 \text{ min} \\ \tau_2 &= 0.1667 \text{ min} \\ \theta &= 0.35 \text{ min}\end{aligned}$$

Using Skogestad's approximation, the reduced constants became

$$K_p = 0.333$$

$$\begin{aligned}\tau &= \tau_1 + \frac{\tau_2}{2} \\ \tau &= 1.0833 \text{ min} \\ \theta^* &= \theta + \frac{\tau_2}{2} \\ \theta^* &= 0.4333 \text{ min}\end{aligned}$$

The corresponding first-order approximation was

$$G_p(s) \approx \frac{0.333e^{-0.4333s}}{1.0833s + 1}$$

This simplified model was used only to calculate controller parameters. All controllers were later evaluated using the full process model to ensure a fair comparison.

### 3.3 Calculation of Controller Constants

The controller constants for each tuning method were calculated using the appropriate relations associated with that method. Although the calculation procedures differed, all controllers were ultimately evaluated on the same process model to ensure that performance differences reflected the effect of controller tuning rather than differences in plant representation.

The controller developed in Section 2 was retained without modification. It was implemented as a PI controller because PI control is a natural choice for level-control processes and provides sufficient flexibility to satisfy the required robustness conditions.

#### Ziegler–Nichols Open Loop Controller

The controller constants for the Ziegler–Nichols Open Loop method were calculated using empirical relations derived from the first-order-plus-time-delay model.

$$\begin{aligned}K_c &= \frac{1.2}{\alpha K_p} \frac{\tau}{\theta} \\ \tau_I &= \frac{2\theta}{\theta} \\ \tau_D &= \frac{\theta}{2}\end{aligned}$$

where

$$\alpha = 1$$

was used as the safety factor, corresponding to the original Ziegler–Nichols correlation without additional conservatism.

### Cohen–Coon Controller

The controller constants for the Cohen–Coon method were calculated using the following relations:

$$\begin{aligned}K_c &= \frac{1.35}{\alpha K_p} \left( \frac{\tau}{\theta} + 0.185 \right) \\ \tau_I &= 2.5\theta \left( \frac{\tau + 0.185\theta}{\tau + 0.611\theta} \right) \\ \tau_D &= 0.37 \left( \frac{\tau\theta}{\tau + 0.185\theta} \right)\end{aligned}$$

Again, a value of

$$\alpha = 1$$

was used.

### IMC Controller

The controller constants for the Internal Model Control method were calculated using the direct synthesis tuning relations.

$$\begin{aligned}K_c &= \frac{\tau + \theta/2}{K_p(\tau_c + \theta/2)} \\ \tau_I &= \tau + \frac{\theta}{2} \\ \tau_D &= \frac{\tau\theta/2}{\tau + \theta/2}\end{aligned}$$

where

$$\tau_c = \text{controller tuning parameter}$$

This method typically produces smoother responses and improved robustness compared with aggressive empirical tuning rules.

### Ziegler–Nichols Closed Loop Controller

The controller constants for the closed-loop Ziegler–Nichols method were obtained using the ultimate gain and ultimate period.

$$\begin{aligned}K_c &= 0.60K_u \\ \tau_I &= 0.50P_u \\ \tau_D &= 0.125P_u\end{aligned}$$

### **Tyreus–Luyben Controller**

The Tyreus–Luyben controller constants were obtained from more conservative tuning relations:

$$\begin{aligned}K_c &= 0.45K_u \\ \tau_I &= 2.2P_u \\ \tau_D &= 0.1587P_u\end{aligned}$$

These relations produce slower but more stable responses compared with the Ziegler–Nichols closed-loop method.

### **Personal Controller**

The personal controller was selected directly rather than from a published tuning correlation. The selected parameters were

$$\begin{aligned}K_c &= 4.0 \\ \tau_I &= 1.2 \text{ min} \\ \tau_D &= 0\end{aligned}$$

This controller was chosen to provide a smooth and well-damped response without derivative action.

## **3.4 Time-Domain Evaluation Using the Automation Panel**

After the controller constants were determined, each controller was evaluated using time-domain simulation. The same plant model was retained for every controller to ensure that the comparison reflected differences in controller tuning rather than differences in system representation.

The process model used for evaluation was

$$G_p(s) = \frac{0.333e^{-0.35s}}{(1.0s + 1)(0.1667s + 1)}$$

For each controller, two responses were generated:

1. Unit set-point change
2. Unit disturbance change

The resulting responses were used to evaluate transient performance.

The numerical performance measures recorded from the simulation included:

- Overshoot
- Settling time

These values were taken directly from the simulation output and summarized in the comparison table.

It is important to note that disturbance responses may include negative deviations before returning to steady state. For this reason, disturbance settling time was considered a more reliable indicator of performance than overshoot alone.

### **3.5 Robustness Analysis Using the Bode Stability Sheet**

In addition to time-domain performance, each controller was evaluated in the frequency domain to assess robustness. Robustness describes the ability of the control system to remain stable when system parameters change or when additional time delay is introduced.

The robustness measures considered in this analysis included:

- Gain margin
- Phase margin
- Maximum allowable additional delay
- Peak sensitivity
- Peak complementary sensitivity

#### **Gain Margin**

The gain margin was calculated using the amplitude ratio at the critical frequency:

$$GM = \frac{1}{AR(\omega_c)}$$

A larger gain margin indicates that the system can tolerate greater increases in loop gain before becoming unstable.

#### **Phase Margin**

The phase margin was evaluated at the gain crossover frequency:

$$PM = 180^\circ + PA(\omega_g)$$

A larger phase margin corresponds to improved stability and reduced oscillation.

### Maximum Allowable Additional Delay

The maximum additional delay that the system can tolerate before instability occurs was calculated using

$$\Delta\theta_{max} = \frac{PM}{\omega_g} \frac{\pi}{180}$$

This value represents the additional time delay that can be introduced without causing instability.

### Peak Sensitivity

The peak sensitivity was calculated as

$$M_S = \max \left[ \frac{1}{|1 + G_c(j\omega)G_p(j\omega)|} \right]$$

This value indicates how close the system is to instability. Lower values correspond to better robustness.

### Peak Complementary Sensitivity

The peak complementary sensitivity was calculated as

$$M_T = \max \left[ \frac{|G_c(j\omega)G_p(j\omega)|}{|1 + G_c(j\omega)G_p(j\omega)|} \right]$$

Lower values indicate less aggressive controller behavior.

## 3.6 Overall Comparison of All Controllers

The comparison of time-domain performance and robustness revealed consistent trends among the different tuning methods. Each controller exhibited distinct behavior depending on how aggressively it was tuned and how sensitive it was to changes in system conditions.

The aggressive controllers, namely the Cohen–Coon and Ziegler–Nichols Open Loop methods, produced fast responses but also generated significant oscillations. Their robustness values were comparatively weak, indicating that these controllers would be more sensitive to disturbances or modeling uncertainty.

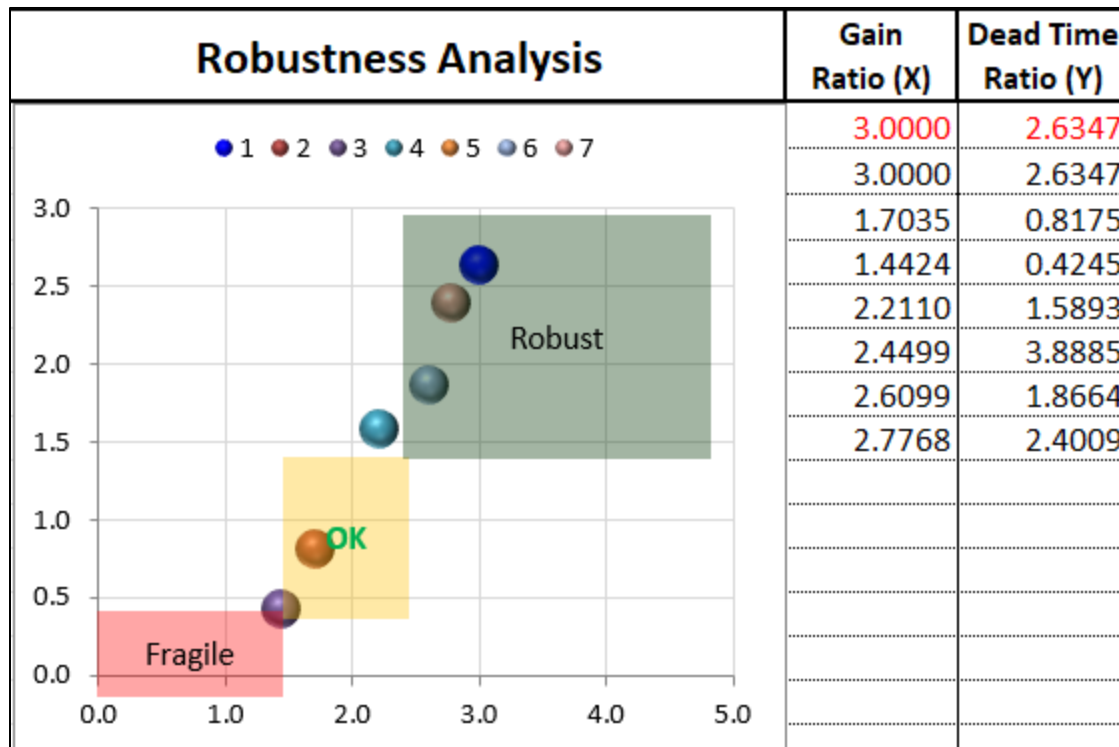


Figure 3- Robustness Analysis

The conservative controller, Tyreus–Luyben, demonstrated excellent robustness. Its gain margin and phase margin values were significantly larger than those of the other controllers. However, this increased stability came at the expense of slower response time.

The most balanced controllers were the IMC controller, the personal controller, and the GM = 3, PM = 60° controller. These controllers produced smooth responses with moderate settling times while maintaining acceptable robustness.

Table 1- Controller performance comparison

|                                                                           |             | GM=3<br>PM=60 deg | Ziegler- Nichols<br>Open Loop | Cohen-Coon | Ziegler-Nichols<br>Closed Loop | Luyben-Tyreus | IMC   | My Personal<br>Controller | Selected Controller<br>IMC |
|---------------------------------------------------------------------------|-------------|-------------------|-------------------------------|------------|--------------------------------|---------------|-------|---------------------------|----------------------------|
| Overshoot (if present)                                                    | Set-point   | 0.07              | 0.48                          | 0.70       | 0.18                           | 0.00          | 0.10  | 0.08                      | 0.10                       |
|                                                                           | Disturbance | 0.16              | 0.11                          | 0.11       | 0.11                           | 0.12          | 0.12  | 0.16                      | 0.12                       |
| Settling time, min                                                        | Set-point   | 3.90              | 3.02                          | 5.48       | 3.64                           | 17.54         | 3.01  | 4.03                      | 3.01                       |
|                                                                           | Disturbance | 3.64              | 1.79                          | 2.92       | 2.71                           | 13.04         | 3.42  | 3.63                      | 3.42                       |
| Gain Margin                                                               |             | 3.00              | 1.70                          | 1.44       | 2.21                           | 2.45          | 2.61  | 2.78                      | 2.61                       |
| Phase Margin, deg                                                         |             | 60.00             | 42.44                         | 26.36      | 58.88                          | 101.98        | 62.72 | 58.93                     | 62.72                      |
| Maximum amount of time delay allowed before the onset of instability, min |             | 0.92              | 0.29                          | 0.15       | 0.56                           | 1.36          | 0.65  | 0.84                      | 0.65                       |
| Peak Value of the sensitivity function                                    |             | 1.67              | 2.56                          | 3.76       | 1.88                           | 1.72          | 1.72  | 1.74                      | 1.72                       |
| Peak Value of the complimentary sensitivity function                      |             | 1.00              | 1.68                          | 2.96       | 1.07                           | 1.00          | 1.00  | 1.04                      | 1.00                       |
| Controller Gain, Kc                                                       |             | 3.65              | 9.01                          | 10.88      | 6.57                           | 4.92          | 6.01  | 4.00                      | 6.01                       |
| Controller Reset Time, $\tau_i$                                           |             | 1.132             | 0.867                         | 0.997      | 0.985                          | 4.335         | 1.30  | 1.20                      | 1.30                       |
| Controller Derivative Time, $\tau_d$                                      |             | 0.000             | 0.217                         | 0.149      | 0.246                          | 0.313         | 0.181 | 0.000                     | 0.181                      |

### 3.7 Final Controller Selection and Ranking

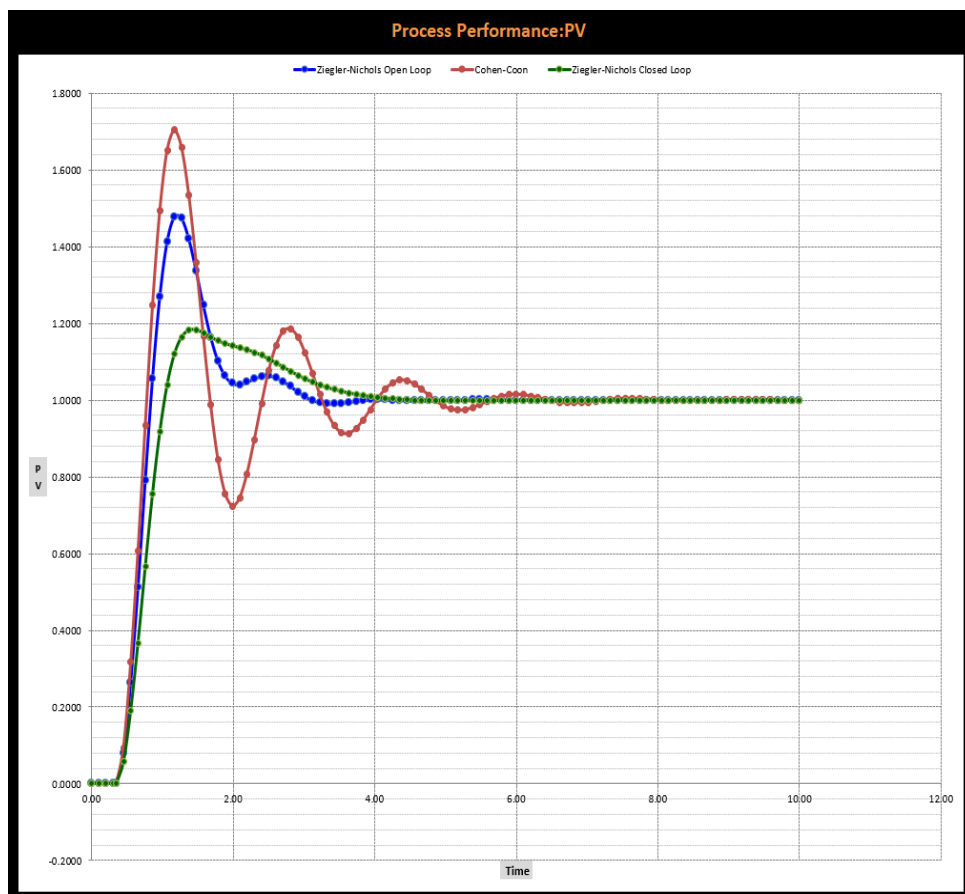
Based on the complete comparison of performance and robustness, the IMC controller was selected as the final recommended controller for the two-tank level-control system.

This controller provided the best compromise between response speed and stability. Its set-point response was smooth and well damped, and its disturbance rejection performance remained acceptable. In addition, its robustness values indicated reliable operation under changing process conditions.

The controllers were ranked from best to worst as follows:

1. IMC Controller
2. Personal Controller
3.  $GM = 3$ ,  $PM = 60^\circ$  Controller
4. Tyreus–Luyben Controller
5. Ziegler–Nichols Closed Loop Controller
6. Ziegler–Nichols Open Loop Controller
7. Cohen–Coon Controller

The figures below shows the performance of the controller with setpoint or disturbance changes:



*Figure 4- Change in Setpoint ZN-Open loop, Cohen-Coon, and ZN-Closed Loop Controllers*

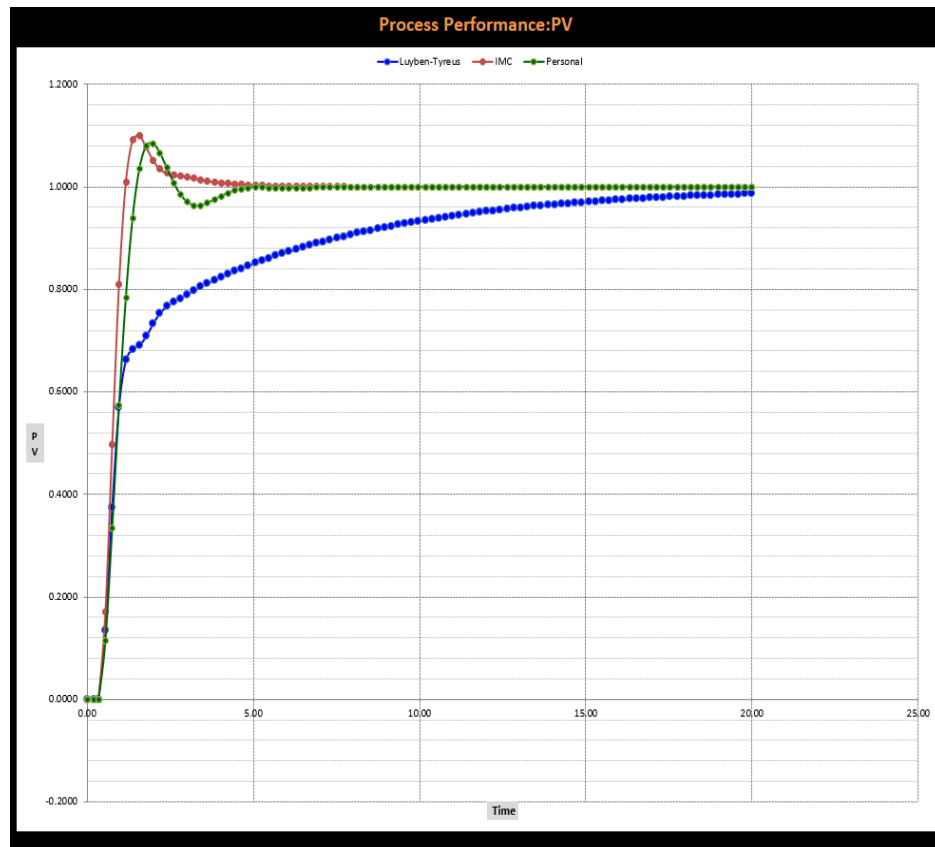
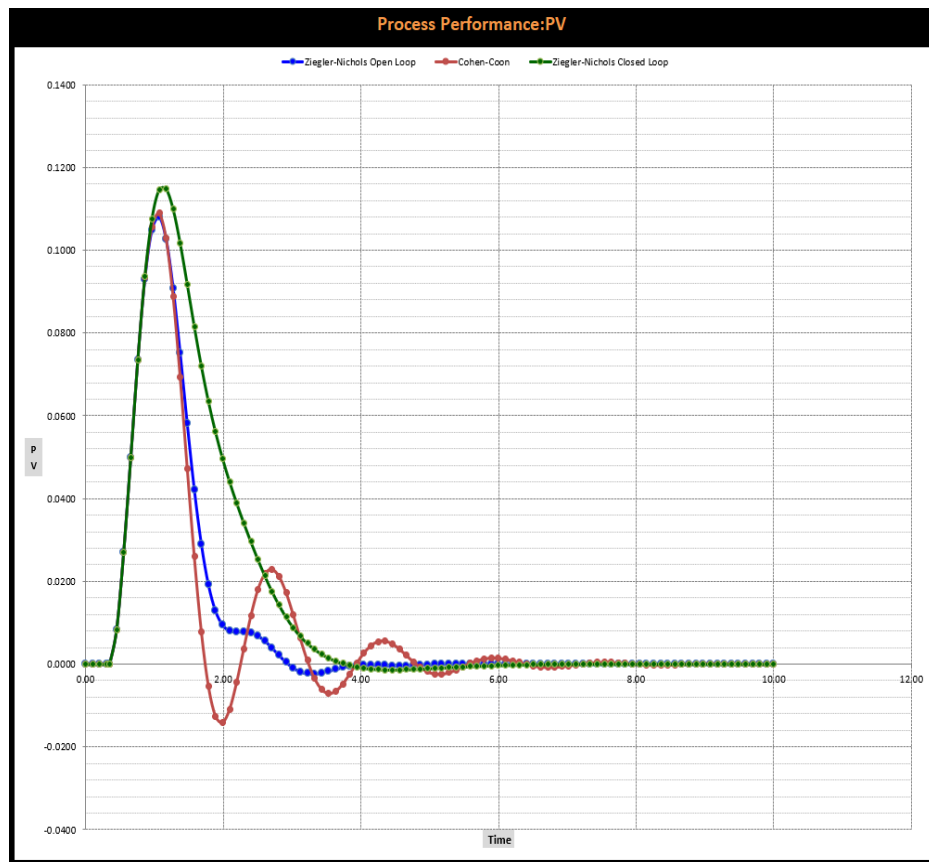


Figure 5- Change in Setpoint for Luyben-Tyresus, IMC, and Personal Controllers



*Figure 6- Change in disturbance for ZN-Open loop, Cohen-Coon, and ZN-Closed Loop Controllers*

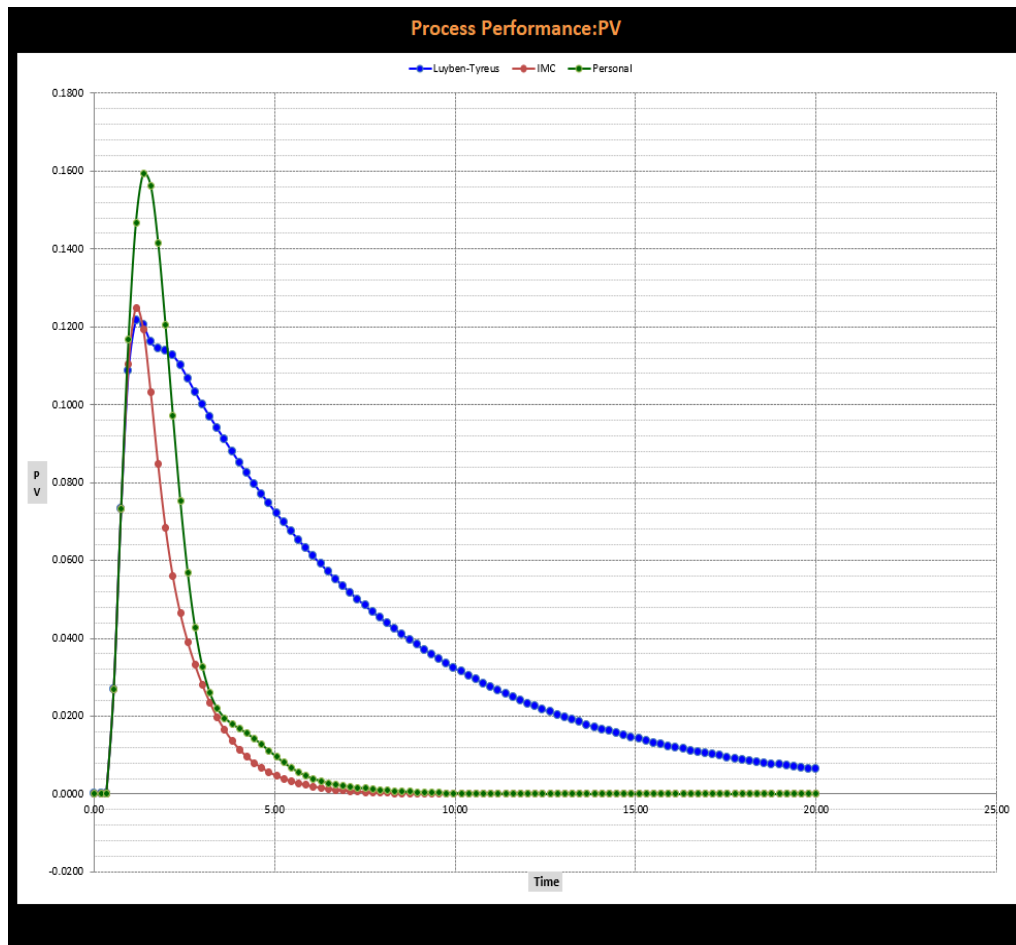


Figure 7- Change in disturbance Luyben-Tyres, IMC, and Personal Controllers

## Conclusion

This assignment demonstrated the importance of proper controller tuning in maintaining stable and efficient operation of dynamic processes. The interacting two-tank system was modeled using dynamic mass balances, and the resulting transfer function showed that the system behaves as a stable second-order process with time delay.

Several controller tuning methods were evaluated and compared using both time-domain performance and frequency-domain robustness analysis. The results showed that aggressive tuning methods produced faster responses but reduced stability, while conservative methods improved robustness but slowed system performance.

Among all the controllers tested, the Internal Model Control method provided the best balance between performance and robustness. For this reason, the IMC controller was selected as the final recommended controller for the two-tank level-control system.

## Appendix- Code

Function FindOmegaAtAR(omegaVals As Range, arVals As Range, targetAR As Double) As Double

Dim rowNum As Long

Dim omegaLow As Double, omegaHigh As Double

Dim arLow As Double, arHigh As Double

For rowNum = 2 To arVals.Rows.Count

arLow = arVals.Cells(rowNum - 1, 1).Value

arHigh = arVals.Cells(rowNum, 1).Value

If (arLow >= targetAR And arHigh <= targetAR) Or \_

(arLow <= targetAR And arHigh >= targetAR) Then

omegaLow = omegaVals.Cells(rowNum - 1, 1).Value

omegaHigh = omegaVals.Cells(rowNum, 1).Value

If arHigh = arLow Then

FindOmegaAtAR = omegaLow

Else

FindOmegaAtAR = omegaLow + \_

(targetAR - arLow) \* (omegaHigh - omegaLow) / (arHigh - arLow)

End If

Exit Function

End If

Next rowNum

FindOmegaAtAR = CErr(xlErrNA)

End Function

Function FindOmegaAtPhase(omegaVals As Range, phaseVals As Range, targetPhase As Double) As Double

Dim rowNum As Long

Dim omegaLow As Double, omegaHigh As Double

Dim phaseLow As Double, phaseHigh As Double

For rowNum = 2 To phaseVals.Rows.Count

phaseLow = phaseVals.Cells(rowNum - 1, 1).Value

phaseHigh = phaseVals.Cells(rowNum, 1).Value

If (phaseLow >= targetPhase And phaseHigh <= targetPhase) Or \_

(phaseLow <= targetPhase And phaseHigh >= targetPhase) Then

omegaLow = omegaVals.Cells(rowNum - 1, 1).Value

omegaHigh = omegaVals.Cells(rowNum, 1).Value

If phaseHigh = phaseLow Then

```

        FindOmegaAtPhase = omegaLow
    Else
        FindOmegaAtPhase = omegaLow + _
            (targetPhase - phaseLow) * (omegaHigh - omegaLow) / (phaseHigh -
phaseLow)
    End If
    Exit Function
End If
Next rowNum

FindOmegaAtPhase = CVErr(xlErrNA)
End Function

```

## References

- 1- Caracotsios, M. (2026). Introduction to chemical process control (Chapter 4) [Lecture notes]. Department of Chemical Engineering, University of Illinois Chicago.
- 2- Caracotsios, M. (2026). Introduction to chemical process control (Chapter 5) [Lecture notes]. Department of Chemical Engineering, University of Illinois Chicago.
- 3- Caracotsios, M. C. (2023). CHE Calculator (Version 14) [Computer software]. Department of Chemical Engineering, University of Illinois Chicago.
- 4- Caracotsios, M. (2026). Introduction to chemical process control (Chapter 6) [Lecture notes]. Department of Chemical Engineering, University of Illinois Chicago.